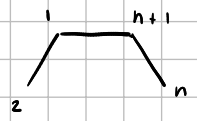


observation ①

$X_{2\,n+1} = -X_{13}$

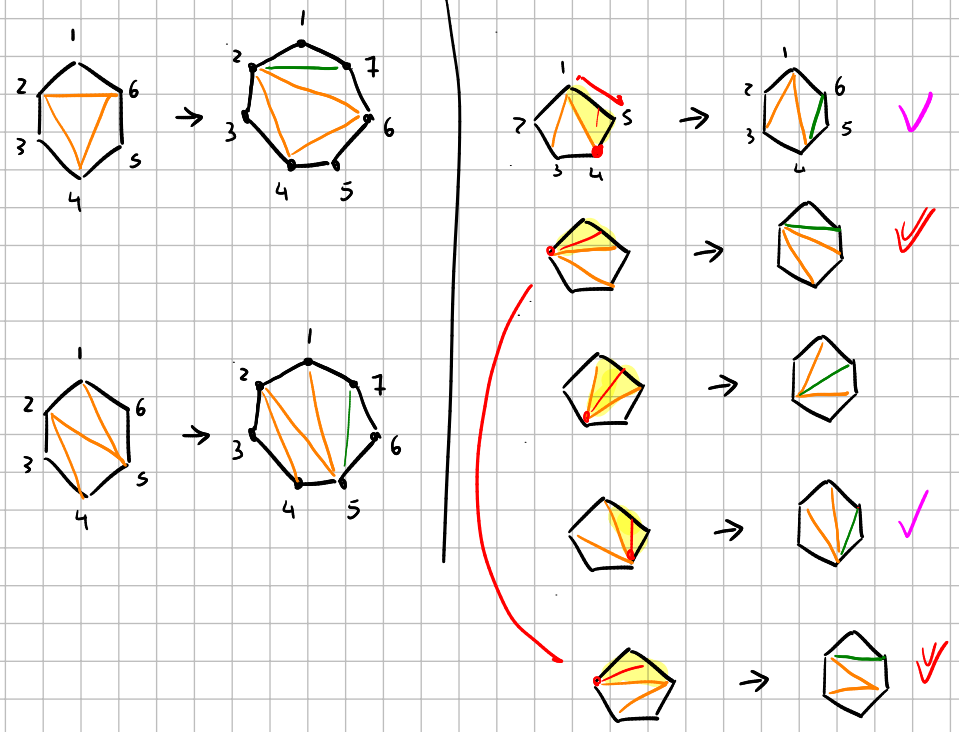
$X_{2n} = X_{1n} - X_{13}$

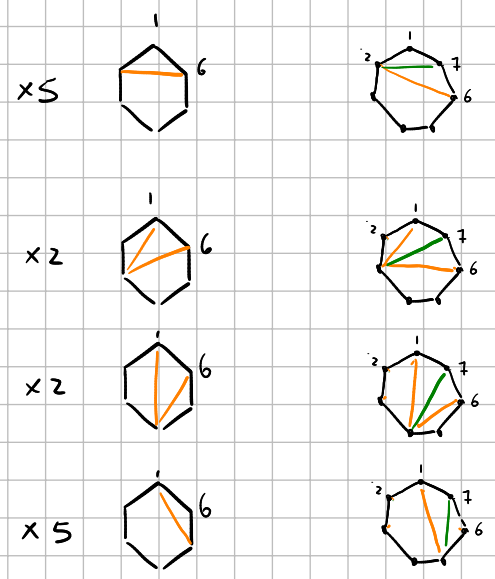
$$\left.\begin{array}{l} X_{3\,n+1} \\ X_{4\,n+1} \\ \vdots \\ X_{n-1\,n+1} \end{array}\right\} \text{new free variables}$$

new chords are either in new equations or new free variables

observation ②

ex





group non-bad things in n+1

- non bad terms made of chords that existed in n-gon
- triangulation in n-gon and add an intersecting chord (starting from vertex n+1)
- more than one new chord that break triangulation

$$B = \sum_{i_1 j_1} \sum_{i_2 j_2} \frac{a_{i_1 j_1 i_2 j_2}}{x_{i_1 j_1} x_{i_2 j_2}}$$

$x_{24} = x_{14} - x_{13}$

$x_{25} = -x_{13}$

x_{35} arbitrary

Since x_{35} arbitrary: $\frac{a_{35\,35}}{x_{35}^2} = 0$

$a_{35\,35} = 0$

- $x_{13} \to \infty$ x_{14}, x_{35} arbitrary $x_{24}, x_{25} \to \infty$ $\quad 0 = \frac{a_{14\,35}}{x_{14}\, x_{35}}$ $\quad a_{14\,35} = 0$
- $x_{13} \to \infty$ x_{24}, x_{35} arbitrary $x_{14}, x_{25} \to \infty$ $\quad 0 = \frac{a_{24\,35}}{x_{24}\, x_{35}}$ $\quad a_{24\,35} = 0$

- x_{13} fixed x_{35} fixed $x_{25} = -x_{13}$ $\quad x_{14} \to \infty$, $x_{24} \to \infty$

$$\frac{a_{13\,35}}{x_{13}\, x_{35}} - \frac{a_{25}\, a_{35}}{x_{13}\, x_{35}} = 0 \qquad a_{13\,35} = a_{25\,35}$$

$$\frac{a_{13\,13}}{x_{13}^2} - \frac{a_{13\,25}}{x_{13}^2} = 0 \qquad a_{13\,13} = a_{13\,25}$$

$$\frac{a_{25\,25}}{x_{13}^2} - \frac{a_{13\,25}}{x_{13}^2} = 0 \qquad a_{25\,25} = a_{13\,25}$$

cyclic permutations

$a_{25}\, a_{46} = 0$

$\sum_{\text{bad terms}}$ satisfies 1-zero

$(n=5)$

$$A = \frac{a}{x_{13} x_{35}} + \frac{a}{x_{24} x_{41}} + \frac{a}{x_{35} x_{52}} + \frac{a}{x_{41} x_{13}} + \frac{a}{x_{52} x_{24}}$$

$$B = A + a\left(\frac{1}{\cdots} + \frac{1}{\cdots} + \frac{1}{\cdots} + \cdots\right)$$

actual amplitude $\subseteq$ most general one

$$\sum_{i,k,l} \frac{a_{ikl}}{x_{ij}\, x_{kl}}$$

$S = \{$ all possible potential amplitudes $\}$ that satisfies the 1-zeros

A actual amplitude $\in S$

if we prove that $S = \{ a\, C \mid a \in \mathbb{R} \}$

$$\begin{cases} y'(t) = y(t) \\ y(0) = y_0 \end{cases} \qquad y = y_0 e^t$$

$N = n-3$

$$B = \sum_{\substack{\{i_1 j_1\} \\ \{i_2 j_2\} \\ \vdots \\ \{i_N j_N\}}} \frac{a_{i_1 j_1 i_2 j_2 \cdots i_N j_N}}{X_{i_1 j_1} \cdots X_{i_N j_N}}$$

$$\begin{cases} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ \vdots \\ X_{2\,n-1} = X_{1\,n-1} - X_{13} \\ X_{2n} = -X_{13} \end{cases}$$

X_{13} X_{14} $X_{1\,n-2}$

X_{24} X_{25} $X_{2\,n-1}$

X_{35} X_{36} $\cdots$ X_{3n}

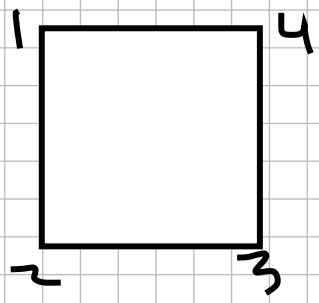
X_{46} $\cdots$ X_{4n}

$\ddots$

$X_{n-2\,n}$

$X_{n-1\,1}$

X_{2n}



n =4. we only pick 1 chord. so we force to pick 13 or 24. if 13 = triangulation. if 24 = triangulation.

B = a13/X_13 + a24/X_24

since X24 = -X13

B = (a13 - a24)/X13 ->

a_13=a_24

+1

$$\begin{cases} X_{35} = X_{25} - X_{24} \\ \vdots \\ X_{3n} = X_{2n} - X_{24} \\ X_{31} = -X_{24} \end{cases}$$

X_{46} X_{47} $\cdots$ X_{41}

X_{57} $\cdots$ X_{51}

$X_{1\,n-1}$

- for each X_{ij} there is a 1-zero where X_{ij} is a free variable

send everything but X_{ij} to ∞ $\qquad 0 = \dfrac{\boxed{a_{i_1 j_1 \cdots i_N j_N}}}{X_{ij}^{N}} = 0$

<u>things that can be arbitrary</u> on top of

X_{35} X_{36} $\cdots$ X_{3n}

X_{46} $\cdots$ X_{4n}

$\ddots$

$X_{n-2\,n}$

?

$$\begin{cases} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ \vdots \\ X_{2\,n-1} = X_{1\,n-1} - X_{13} \\ X_{2n} = -X_{13} \end{cases}$$

$\to -\infty$ $\qquad \to \infty$

- choose one from each pair (X_{14}, X_{24}) (X_{15}, X_{25}), $\ldots$ $(X_{2n-1}, X_{1\,n-1})$

all of the free variables are arbitrary. rest goes to ∞

only terms that have these arbitrary variables survive (their total power must be N)

since now a polynomial in these variables is zero, all the coefficients must be zero

---

$n=5$

$$\frac{a_{13\,13}}{x_{13}^2} + \boxed{\frac{a_{13\,14}}{x_{13}x_{14}}} + \frac{a_{13\,24}}{x_{13}x_{24}} + \frac{a_{13\,25}}{x_{13}x_{25}} + \boxed{\frac{a_{13\,35}}{x_{13}x_{35}}}$$

$i \to i+1$

$j \to j+1$

$$\frac{a_{24\,24}}{x_{24}^2} + \frac{a_{24\,25}}{x_{24}x_{25}} + \frac{\cdot\cdot}{x_{24}x_{35}} + \frac{\cdot\cdot}{x_{24}x_{31}} + \boxed{\frac{\cdot\cdot}{x_{24}x_{41}}}$$

$$\frac{\cdot\cdot}{x_{35}^2} + \frac{\cdot\cdot}{x_{35}x_{31}} + \frac{\cdot\cdot}{x_{35}x_{41}} + \frac{\cdot\cdot}{x_{35}x_{42}} + \boxed{\frac{\cdot\cdot}{x_{35}x_{52}}}$$

$$\frac{\cdot\cdot}{x_{41}^2} + \frac{\cdot\cdot}{x_{41}x_{42}} + \boxed{\frac{\cdot\cdot}{x_{41}x_{52}}} + \frac{\cdot\cdot}{x_{41}x_{53}} + \boxed{\frac{\cdot\cdot}{x_{41}x_{13}}}$$

$$\frac{\cdot\cdot}{x_{52}^2} + \frac{\cdot\cdot}{x_{52}x_{53}} + \frac{\cdot\cdot}{x_{52}x_{13}} + \frac{\cdot\cdot}{x_{52}x_{14}} + \boxed{\frac{\cdot\cdot}{x_{52}x_{24}}}$$

$x_{13}\ x_{14}$
$x_{24}\ x_{25}$
x_{35}

$a_{14\,13} = a_{14\,25}$

$a_{35\,13} = a_{35\,25}$

$$\begin{cases} x_{24} = x_{14} - x_{13} \\ x_{25} = -x_{13} \end{cases}$$

x_{35} arbitrary

sets $a_{\cdot\cdot} = 0$

step 1 → choose 2 arbitrary variables, send rest to ∞

- x_{35}, x_{14} $\quad \frac{\quad}{x_{35}^2} \quad \frac{\quad}{x_{14}^2} \quad \frac{\quad}{x_{14}x_{35}} \rightarrow$ $a_{35\,35} = 0$, $a_{14\,14} = 0$, $a_{14\,35} = 0$

- x_{35}, x_{24} $\quad \frac{\quad}{x_{35}^2} \quad \frac{\quad}{x_{24}^2} \quad \frac{\quad}{x_{24}x_{35}} \rightarrow$ $a_{24\,35} = 0$, $a_{24\,24} = 0$

---

step 2 → sets $a_{\cdot\cdot} = a_{\cdot\cdot}$

chose x_{13} arbitrary, x_{35} arbitrary $\quad \boxed{x_{25} = -x_{13}}$

chose one of x_{14}, x_{24} arbitrary

- X_{13} X_{35} X_{14} arbitrary $X_{25} = -X_{13}$

$X_{24} \to \infty$

what remains is a polynomial in X_{13}, X_{35}, X_{14}

all coefficient have to be zero

$$\frac{1}{X_{13}^2}\left(\cancel{a_{1313}}^{\,0} + \cancel{a_{2525}}^{\,0} - a_{1325}\right) = 0 \qquad a_{1325} = 0$$

$$\frac{1}{X_{14}X_{13}}\left(a_{1413} - a_{1425}\right) = 0 \qquad a_{1413} = a_{1425}$$

$$\frac{1}{X_{35}X_{13}}\left(a_{3513} - a_{3525}\right) = 0 \qquad a_{3513} = a_{3525}$$

every time we kill something, we kill entire column

$n = 6$

$$\begin{cases} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ X_{26} = -X_{13} \end{cases}$$

X_{35} X_{36} X_{46} arbitrary

35	36	46
46	41	51
51	52	62
62	63	13
13	14	24

Step 1

X_{35} X_{14} X_{15} arbitrary

X_{36} X_{43}

etc ...

$a_{35\,35\,35} = 0$

$a_{35\,35\,14} = 0$ etc

$\vdots$

$\boxed{a_{35\,35\,46} = 0}$

Step 2

X_{35} X_{36} X_{45} X_{13} arbitrary

$X_{26} = -X_{13}$ set to ∞

from step 1 + cyclic shifts

remains:

$$\frac{1}{X_{13}^3}\left(\cancel{a_{1313}}^{\,0} - \underbrace{a_{1313\,26}}_{=0} + \underbrace{a_{1326\,26}}_{=0} - \cancel{a_{2626\,26}}^{\,0}\right)$$

$a_{35\,35\,46} = 0$

$$\frac{1}{X_{13}^2 X_{35}}\left(a_{351313} + a_{352626} - a_{351326}\right) = 0$$

$$\frac{1}{X_{35}^2 X_{13}}\left(a_{353513} - a_{353526}\right) = 0$$

$\to$ only 1 power is resist

Very possible that from step 1

we can always shift so that 13 and 2n are now arbitrary

$a_{i_1 i_2 \cdots 13} = a_{i_1 i_2 \cdots 26}$

try with Wolfram